

Applied Thermo-fluids I

3-0-0

Pipe Networks- Moody's diagram, pipe network, calculations for flow rate-head loss-pumping power, hydraulic diameter of ducts [5 hours]

Ideal Flows- Potential flow analysis, elementary flows, flow over cylinders, lift and drag, Magnus effect [7 hours]

Hydraulic Machines– Euler pump/turbine equation, classification of hydraulic machines, Impulse momentum principle, velocity triangles, efficiency, Centrifugal and Axial pumps, velocity triangles and analysis, effect of blade angle, cavitation, NPSH, priming and testing of pumps, specific speed, characteristic curves, Pelton, Francis and Kaplan turbines, specific speed, efficiency [12 hours]

Compressible Flow– Regimes of compressible flow, Mach number, Adiabatic and isentropic steady flow, governing equations for 1 –D inviscid flows, stagnation properties, Isentropic flow with area change, convergent, divergent and convergent-divergent nozzles, Rankine-Hugoniot relations and introduction to shocks, oblique shock and expansion wave [12 hours]

Text Books: 1. Fluid Mechanics- Fox, McDonald, Pritchard, 8th Edition, Wiley 2. Introduction to Fluid Mechanics and Fluid Machines- Som, Biswas and Chakraborty, 3rd edition, McGraw-Hill 3. Fluid Mechanics- F M White, 8th Edition, McGraw-Hill

Reference Books: 1. J. D. Anderson (Jr.), Modern Compressible Flow, McGraw-Hill , 2. S. L. Dixon, and C. A. Hall, Fluid Mechanics and Thermodynamics of Turbomachinery, Elsevier.